

Access: <https://zenodo.org/records/22144053>

## **The Chaotic Behaviour of Geodesics in Non-homogeneous Vacuum pp-Wave Solutions**

Master of Science in Physics

Famagusta, Cyprus

18 September 2000

- Thesis uses numerical and asymptotic methods to investigate nonlinear behaviour in astrophysics.
- Appendix: Maple V Source Codes.

### **Abstract**

We demonstrate chaotic behaviour of time-like, space like and null geodesics of free test particle in non-homogeneous vacuum plane fronted gravitational waves with asymptotic analytical and numerical methods. Geodesic motions that we derived from pp-waves metric, generate deterministic chaotic Hamiltonian flow on plane of real coordinates and corresponding phase-spaces in case of  $n$  greater than or equal to 3, which belongs to famous Henon-Heiles family. Analyzing integrability of this type of Hamiltonian flow, show that even simple and well-known dynamics of pp-waves could evolve in complex fashion.